Solution to Problem 16

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Abstract

We analytically determine the equilibrium shape of a perfectly conducting mercury droplet in a uniform electric field. By balancing hydrostatic pressure, surface tension, and electrostatic forces using a first-order Legendre polynomial expansion, we demonstrate that the droplet forms a prolate spheroid. Explicit expressions for the modified pressure differential and the droplet's eccentricity are derived.

1 Equations Governing the Droplet Geometry

At equilibrium, the electric field inside a perfectly conducting mercury droplet must vanish. Consequently, the internal hydrostatic pressure remains uniform when the sole external influence is an applied electric field. Let $\Delta p \equiv p_{\text{in}} - p_{\text{out}}$ denote the pressure differential across the droplet interface. For an infinitesimal surface element, the forces per unit area arise from the hydrostatic pressure $\Delta p\hat{\boldsymbol{n}}$, the electrostatic pressure $\sigma^2/(2\epsilon_0)\hat{\boldsymbol{n}}$, and the surface tension $-\alpha\nabla\cdot\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}$, where σ is the surface charge density, α is the surface tension coefficient, and $\hat{\boldsymbol{n}}$ is the outward normal vector field. Mechanical equilibrium necessitates that these stress components sum to zero:

$$\Delta p + \frac{\sigma^2}{2\epsilon_0} - \alpha\nabla\cdot\hat{\boldsymbol{n}} = 0. \tag{1}$$

The electrostatic potential $\phi(\boldsymbol{r})$ must satisfy Laplace's equation, subject to appropriate boundary conditions. The resulting potential gradient at the boundary determines the surface charge density σ:

$$\nabla^2\phi = 0, \qquad\qquad \sigma = -\epsilon_0\frac{\partial\phi}{\partial n}. \tag{2}$$

Equations (1) and (2), alongside the constraint of volume conservation,[1] formulate a closed boundary-value problem governing the droplet's geometry, parameterized by the normal field $\hat{\boldsymbol{n}}$ and the surface charge density σ.

[1] The constant-volume constraint is formally addressed in the subsequent analytical derivation.

2 Approximate Analytical Solution

Assuming an applied uniform electric field $\boldsymbol{E}_0$ aligned with the z-axis, azimuthal symmetry dictates that the droplet's surface profile depends exclusively on the polar angle θ. Given the weak-field condition $\epsilon_0 E_0^2 R/\alpha \ll 1$, the structural deformation remains perturbative. We define the surface boundary as $f(r,\theta) = r - R - \epsilon(\theta) = 0$,[2] where R is the unperturbed radius and $|\epsilon(\theta)| \ll R$ represents the angular deformation. Retaining terms up to the first order in ϵ yields the normal vector:

$$\nabla f = \hat{\boldsymbol{r}} - \frac{1}{R}\frac{\mathrm{d}\epsilon}{\mathrm{d}\theta} \to \hat{\boldsymbol{n}} = \frac{\nabla f}{|\nabla f|} \approx \hat{\boldsymbol{r}} - \frac{1}{R}\frac{\mathrm{d}\epsilon}{\mathrm{d}\theta}.$$

Consequently, the divergence of the normal vector evaluates to:

$$\nabla \cdot \hat{\boldsymbol{n}} \approx \frac{2}{R} - \frac{2\epsilon}{R^2} - \frac{1}{R^2 \sin(\theta)}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin(\theta)\frac{\mathrm{d}\epsilon}{\mathrm{d}\theta}\right),$$

where the substitution $r = R + \epsilon$ is applied post-differentiation. Equation (1) can be recast as:

$$\nabla \cdot \hat{\boldsymbol{n}} = \frac{\Delta p}{\alpha} + \frac{\sigma^2}{2\alpha\epsilon_0}.$$

Since the electrostatic pressure term is second-order in the field strength and thus assumed small, we substitute the zeroth-order approximation for the surface charge density of a conducting sphere, $\sigma(\theta) = 3\epsilon_0 E_0 \cos(\theta)$ (see [1], section 2.5). Consolidating these expressions yields a second-order ordinary differential equation for $\epsilon(\theta)$:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\left(1 - x^2\right)\frac{\mathrm{d}\epsilon}{\mathrm{d}x}\right] + 2\epsilon + \frac{9\epsilon_0 E_0^2 R^2}{2\alpha}x^2 + \frac{\Delta p R^2}{\alpha} - 2R = 0, \tag{3}$$

where $x \equiv \cos(\theta)$. To solve this formally, we expand the angular deformation in terms of Legendre polynomials, $P_l(x)$:

$$\epsilon(x) = \sum_{l=0}^{\infty} a_l P_l(x), \quad \text{with} \quad \frac{\mathrm{d}}{\mathrm{d}x}\left[\left(1 - x^2\right)\frac{\mathrm{d}P_l}{\mathrm{d}x}\right] + l(l+1)P_l = 0. \tag{4}$$

Substituting this expansion into Equation (3) yields:

$$\sum_{l=0}^{\infty}\left[2 - l(l+1)\right] a_l P_l(x) + \left(\frac{3\epsilon_0 E_0^2 R^2}{2\alpha} + \frac{\Delta p R^2}{\alpha} - 2R\right) P_0(x) + \frac{3\epsilon_0 E_0^2 R^2}{\alpha} P_2(x) = 0.$$

This imposes the following conditions on the coefficients:

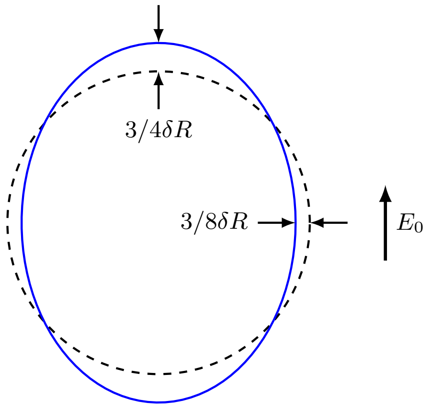
$$2a_0 + \frac{3\epsilon_0 E_0^2 R^2}{2\alpha} + \frac{\Delta p R^2}{\alpha} - 2R = 0, \tag{5a}$$

$$-4a_2 + \frac{3\epsilon_0 E_0^2 R^2}{\alpha} = 0, \tag{5b}$$

$$a_{l>2} = 0, \tag{5c}$$

[2] The overall sign ensures the normal vector correctly points outward from the surface.

Figure 1 First-order deformation of a conducting mercury droplet in a uniform electric field. The dimensionless deformation parameter is defined as $\delta \equiv \epsilon_0 E_0^2 R/\alpha \ll 1$. Under weak-field conditions, the equilibrium geometry approximates a prolate spheroid elongated along the field axis.



leaving a_1 undetermined by the differential equation. To proceed, we impose the volume conservation constraint:

$$\frac{4}{3}\pi R^3 = \frac{1}{3}\int r^3 \,\mathrm{d}\Omega \xrightarrow{r=R+\epsilon} \int \epsilon \,\mathrm{d}\Omega = 0.$$

Utilizing azimuthal symmetry and the Legendre expansion (4), this condition simplifies to:

$$\sum_{l=0}^{\infty} a_l \int_{-1}^{+1} P_l(x)\,\mathrm{d}x = 0. \tag{6}$$

Recognizing the orthogonality of Legendre polynomials ([1], Eq. (3.21)):

$$\int_{-1}^{+1} P_l(x)\,\mathrm{d}x = \int_{-1}^{+1} P_0(x)P_l(x)\,\mathrm{d}x = 2\delta_{0,l},$$

we find that Equation (6) requires $a_0 = 0$. The coefficient a_1 corresponds to a pure translation along the z-axis; by centering the coordinate origin at the droplet's geometric center, we set $a_1 = 0$. Consequently, Equations (5) yield:

$$\Delta p \approx \frac{2\alpha}{R} - \frac{3\epsilon_0 E_0^2}{2}, \tag{7}$$

$$r(\theta) \approx R + \frac{3\epsilon_0 E_0^2 R^2}{8\alpha}\left(3\cos^2(\theta) - 1\right). \tag{8}$$

The presence of the electric field induces an electrostatic pressure that promotes elongation along the field axis, thereby reducing the requisite internal hydrostatic pressure for equilibrium.

Equation (8) analytically defines a prolate spheroid:

$$\frac{z^2}{a^2} + \frac{\rho^2}{b^2} = 1,$$

with semi-major and semi-minor axes given respectively by:

$$a \approx R + \frac{3\epsilon_0 E_0^2 R^2}{4\alpha},$$
$$b \approx R - \frac{3\epsilon_0 E_0^2 R^2}{8\alpha}.$$

The corresponding eccentricity is approximately:

$$e \approx \frac{3}{2}\sqrt{\frac{\epsilon_0 E_0^2 R}{\alpha}}.$$

Figure 1 illustrates this resulting deformation.

References

[1] J. D. Jackson, *Classical electrodynamics*, 3rd ed. (Wiley, New York, 1999).